Adjective Ordering Preferences: An Incremental RSA Account with Bayesian Inference

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Introduction

Speakers exhibit robust preferences in the order of multiple adjectives, preferring expressions like "big brown bag" over "brown big bag". Work on adjective ordering has identified two broad kinds of explanation for this regularity. One emphasizes *stable lexical-semantic asymmetries*: less subjective adjectives are preferred closer to the noun, because they are more reliable under restrictive composition (**???**). The other emphasizes *communicative efficiency in referential context*: adjective order affects how efficiently a listener can identify the intended referent, so speakers should prefer orders that place the most useful discriminative cue early enough to guide reference resolution (**????**).

The present paper argues that these lines of work are best treated as complementary rather than competing. Stable semantic properties such as subjectivity provide a baseline ordering tendency, but actual ordering choices in referential communication are further shaped by the *discriminatory strength* of adjectives in the current context. Our central claim is therefore not that adjective ordering requires a single mechanism such as subjectivity, nor that it requires production-level incrementality in every case. Rather, adjective ordering is a graded solution to a problem of *efficient, incremental reference resolution*: speakers choose an order that balances reliable lexical-semantic information with context-specific pressures on listener search.

To formalize this idea, we develop a Rational Speech Act (RSA) model of adjective ordering with two speaker architectures. A *global* speaker evaluates complete utterances, whereas an *incremental* speaker chooses adjectives left-to-right. Both share a context-sensitive compositional semantics in which adjective meanings are interpreted against evolving comparison classes, allowing us to separate the contribution of compositional semantics from the contribution of online production planning. This comparison is important because recent subjectivity-based theories suggest that incremental semantic restriction may already be sufficient to derive the canonical ordering tendency, while the broader incremental-efficiency literature predicts that online word order should matter whenever it changes discriminatory value.

The project has three main building blocks:

1. **Model.** A Bayesian RSA framework with global and incremental speaker variants, implemented in NumPyro / JAX.
2. **Experiments.** Two complementary referential tasks—a slider-rating experiment and a free-production experiment—that probe both graded preference and actual utterance choice.
3. **Simulation.** A broad parameter sweep that identifies which parts of the ordering effect follow from compositional semantics alone and which are amplified by incremental production.

Discussion: High-level research questions

- Can adjective ordering in referential communication be explained as an efficient solution to incremental reference resolution rather than as a fixed ordering hierarchy?
- How do stable lexical-semantic biases (*e.g.*, subjectivity) interact with context-specific discriminatory strength in shaping ordering preferences?
- What is the minimal mechanism needed for the basic ordering effect, and what additional explanatory value is gained by modeling production as incremental?

1 Background

1.1 Adjective Ordering Beyond Fixed Hierarchies

Adjective ordering preferences have often been treated as evidence for fixed grammatical hierarchies, but recent work has shifted the explanation toward graded cognitive and communicative factors. **?** show that adjective *subjectivity* is a powerful predictor of ordering preferences: less subjective adjectives tend to occur closer to the noun. **?** connect this result to hierarchical nominal composition, arguing that adjectives closer to the noun compose over more restricted comparison classes and therefore benefit from being semantically more reliable. **?** formalize this intuition in simulation, showing that subjectivity-based orderings improve average communicative success. Taken together, this literature suggests that canonical adjective-ordering tendencies can emerge from the interaction of semantic reliability and restrictive composition, without stipulating a rigid ordering template.

1.2 From Subjectivity to Communicative Efficiency

At the same time, a parallel literature in referential communication argues that adjective use and adjective order are shaped by *communicative efficiency* in context. **?** contrasts a subjectivity- or class-based view of ordering with the *discriminatory efficiency hypothesis*, according to which speakers prefer to place the most discriminating adjective earlier in order to facilitate reference resolution. Rubio-Fernández and colleagues extend this idea into a broader *incremental efficiency hypothesis*: the usefulness of a modifier depends not only on whether it is globally informative, but on how it guides listener search as the utterance unfolds word by word (**????**). This literature highlights a distinction that is crucial for the present paper: adjective ordering may reflect not just stable lexical-semantic preferences, but also the *online discriminatory value* of adjectives in a specific referential scene.

1.3 Incremental Pragmatics and the Present Contribution

Incremental RSA models provide a natural way of formalizing this efficiency-based view. In the Cohn-Gordon line of work, pragmatic reasoning is applied to partial utterances rather than only to complete ones, so that the value of a word depends on where it appears and what alternatives remain available (**??**). For adjective ordering, this makes it possible to compare two distinct ideas that the broader literature has often run together: *incremental semantic restriction* on the listener side, and *incremental production planning* on the speaker side.

The current paper contributes at exactly this junction. Empirically, we test whether English adjective-ordering behavior in referential tasks tracks discriminatory strength in context, extending earlier work on subjectivity and referential efficiency. Computationally, we compare a global RSA speaker with an incremental RSA speaker while holding the underlying context-sensitive compositional semantics fixed. This allows us to ask a sharper question than previous work: whether production-level incrementality is necessary for adjective-ordering effects in general, or whether it instead refines a more basic efficiency mechanism grounded in restrictive composition.

Discussion: Positioning the present paper

- **Broader literature connection:** adjective ordering is treated here as part of the broader problem of efficient, incremental reference resolution.
- **Theoretical integration:** subjectivity provides a baseline lexical-semantic tendency; discriminatory strength explains context-sensitive modulation; incremental RSA supplies an explicit process model.
- **Target claim:** we do not argue that only an incremental speaker can generate ordering preferences. Rather, we ask what minimal mechanism derives the baseline effect and whether incremental production improves the account of human behavior.

2 Experiments: Overall Design

The two experiments are complementary. The slider-rating study elicits graded acceptability judgements for pre-specified descriptions, framed from the speaker's perspective (*i.e.*, "how well does this description pick out the target?"), providing continuous data on how well each description fits a given context. The free-production study records the descriptions that speakers actually generate, yielding categorical choice data that reveal ordering preferences and the frequency of redundant modifiers. Together, the two tasks separate semantic fit (*how good is this description?*) from pragmatic choice (*which description does a speaker produce?*).

Both experiments share the same 3×2 factorial design:

Factor	Levels
F_1: **Relevant property**	Size-relevant, Colour-relevant, Both-relevant
F_2: **Sharpness**	Sharp (sizes well separated), Blurred (sizes clustered)
Fixed: Referent = largest, colour-matching, form-matching object in a 6-object context	

The adjective pair shown on each trial varied (size–colour, size–form, or colour–form), but because form and colour share the same semantic type—determinate features parameterised by ν—they are treated identically in the model. All analyses are collapsed across adjective-pair types. Each item encodes a context as a 6×3 matrix (s, c, f) with the referent at index 0 (fig. 1).

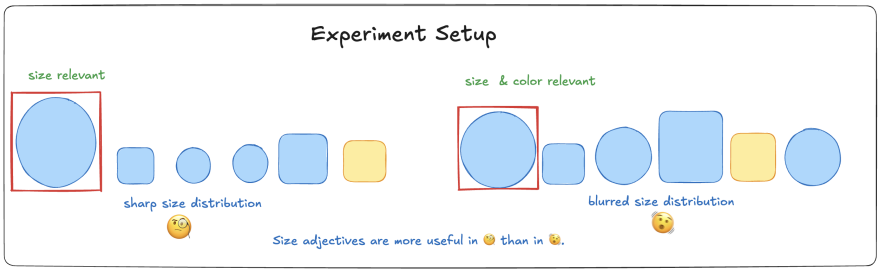

Figure 1: Illustration of an experiment item (placeholder). Left panel: a sharp-size context; right panel: a blurred-size context. The referent (circled) is the largest, colour-matching, form-matching object.

2.1 Slider-rating study (Experiment 1)

Task. Participants rated the *felicity* of each adjective description on a continuous slider (0–100) for a given context-referent pair. Ratings were collected for single-adjective (S, C, F) and two-adjective (SC, SF, CS, *etc.*) descriptions.

Dependent variable. Normalised slider ratings from [0, 100] to [-0.5, 0.5], mean-centred around 0, positive values indicating a preference for size-first orderings and negative values indicating a preference for colour/form-first orderings.

Discussion: Key findings (lmer: centred slider $\sim$ relevance $\times$ sharpness + (1|participant) + (1|item); figs. 2 and 3).

- **Relevance main effect** ($F(2, 3349) = 213.7$, $p < .001$; $\chi^2(2) = 402.6$, $p < .001$): size-first ordering preference is highest in size-relevant contexts, second highest in both-relevant contexts, and lowest in colour-relevant contexts.
- Even in colour-relevant contexts, mean centred ratings remain above zero; the reverse ordering (below zero) is qualitatively not preferred.
- **Sharpness main effect**: not significant ($F(1, 132) = 0.03$, $p = .86$).

- **Relevance × Sharpness interaction** ($F(2, 3349) = 5.84$, $p = .003$; $\chi^2(2) = 11.7$, $p = .003$): sharpness modulates the size-first advantage asymmetrically—it *increases* the preference in size-relevant contexts (blurred: 0.30, sharp: 0.34) but *decreases* it in both-relevant (blurred: 0.25, sharp: 0.20) and colour-relevant contexts (blurred: 0.06, sharp: 0.04).

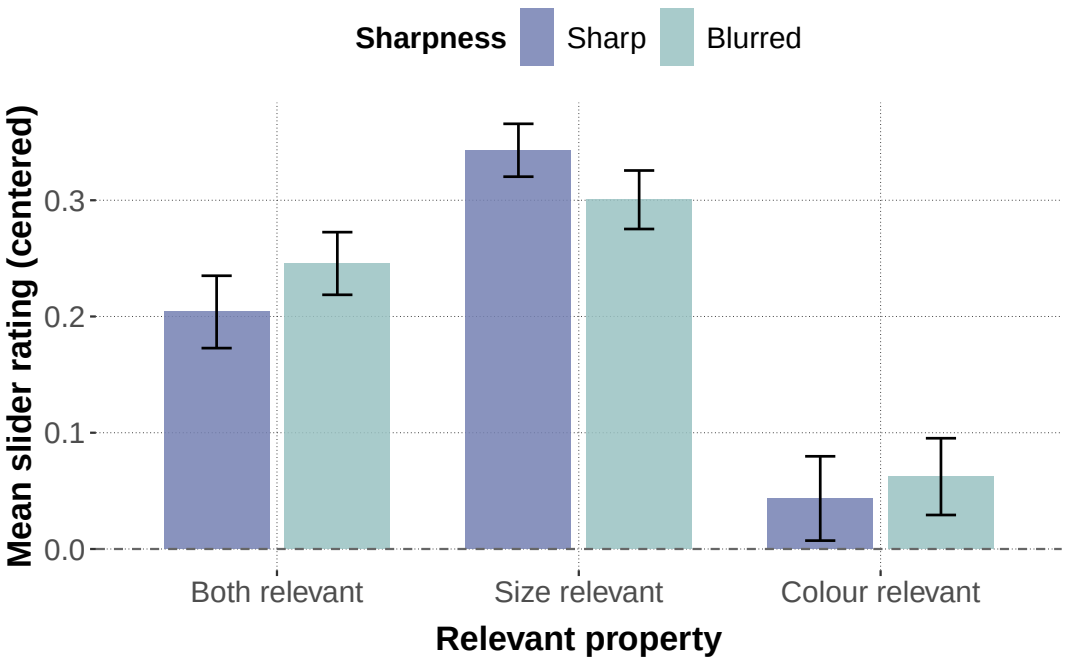

Figure 2: Slider-rating empirical results. Mean centred slider ratings (0 = no preference) by relevant-property condition and sharpness, with 95% CIs. Size-first orderings are preferred across all conditions, but the preference is strongest in size-relevant contexts and weakest in colour-relevant contexts. Sharpness modulates the size-first advantage asymmetrically across relevance conditions.

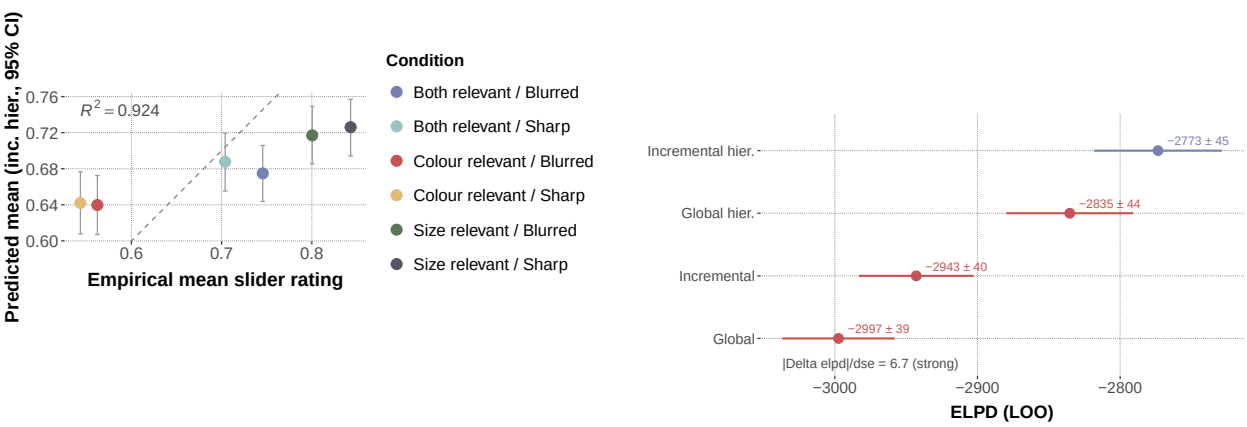

Figure 3: Left: Condition-level observed vs. predicted mean slider ratings for the incremental hierarchical model, with R^2 annotation and 95% CIs. Right: LOO-ELPD comparison across four model variants (best model highlighted in blue); $|\Delta\text{elpd}|/\text{dse} = 6.7$ (strong evidence for the incremental hierarchical model).

2.2 Free-production study (Experiment 2)

Task. Participants described the referent object to a hypothetical listener in as few words as possible, using only the provided adjectives. Responses were manually annotated into the 15 utterance categories.

Dependent variable. Categorical annotation $y_i \in \{1, \ldots, 15\}$ for each trial i.

Discussion: Key findings (glmer: binomial, (1|participant) + (1|item); figs. 4 and 5).

(1) Size-first preference (S-initial utterances coded as 1, all others as 0):

- **Relevance main effect** ($\chi^2(2) = 1859.7$, $p < .001$): size-first orderings dominate in size-relevant contexts; colour-first orderings dominate in colour-relevant contexts; both-relevant falls in between.
- **Sharpness main effect**: not significant ($\chi^2(1) = 0.79$, $p = .37$).
- **Relevance × Sharpness interaction** ($\chi^2(2) = 31.4$, $p < .001$): sharpness modulates the size-first advantage asymmetrically across relevance conditions—it reduces size-first preference in size-relevant contexts but increases it in both-relevant contexts.

(2) Over-informative utterances (three-adjective responses in both-relevant; multi-word responses in size-/colour-relevant coded as 1):

- **Relevance main effect** ($\chi^2(2) = 725.7$, $p < .001$): over-informative utterances are most frequent in size-relevant contexts, less frequent in both-relevant and colour-relevant contexts.
- **Sharpness main effect** ($\chi^2(1) = 10.3$, $p = .001$): over-informative utterances are more frequent in the blurred condition than in the sharp condition.
- **Relevance × Sharpness interaction** ($\chi^2(2) = 63.8$, $p < .001$): the sharpness effect on over-informativity is strongest in size-relevant contexts (blurred elicits far more multi-word utterances) and attenuated in both-relevant and colour-relevant contexts.

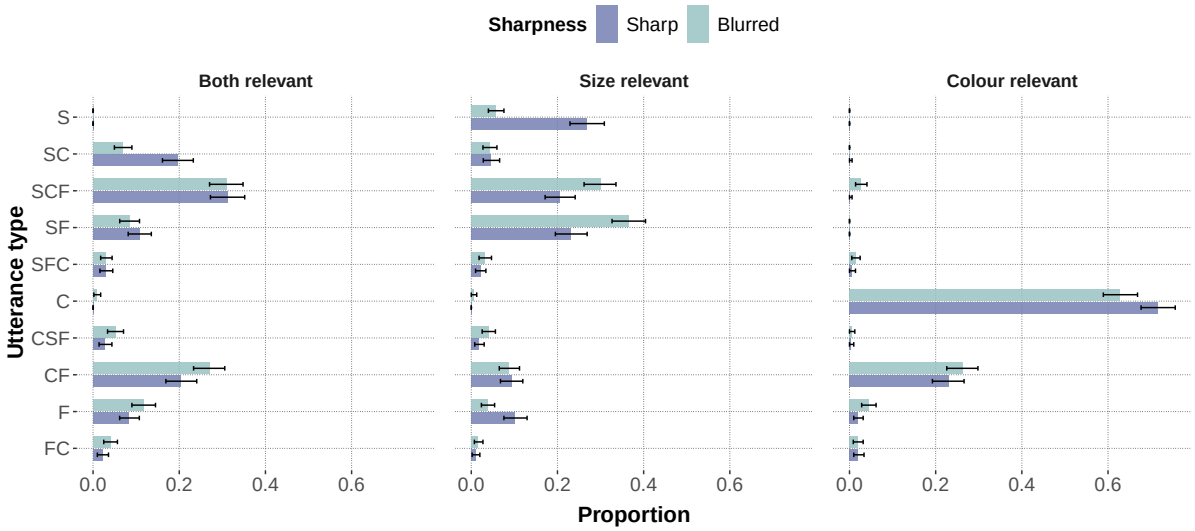

Figure 4: Production empirical results. Utterance-type proportions (with bootstrapped 95% CIs) by relevant-property condition and sharpness. Size-first orderings dominate in size-relevant contexts; colour-first orderings dominate in colour-relevant contexts. Only utterance types with $\geq 2\%$ in at least one condition are shown.

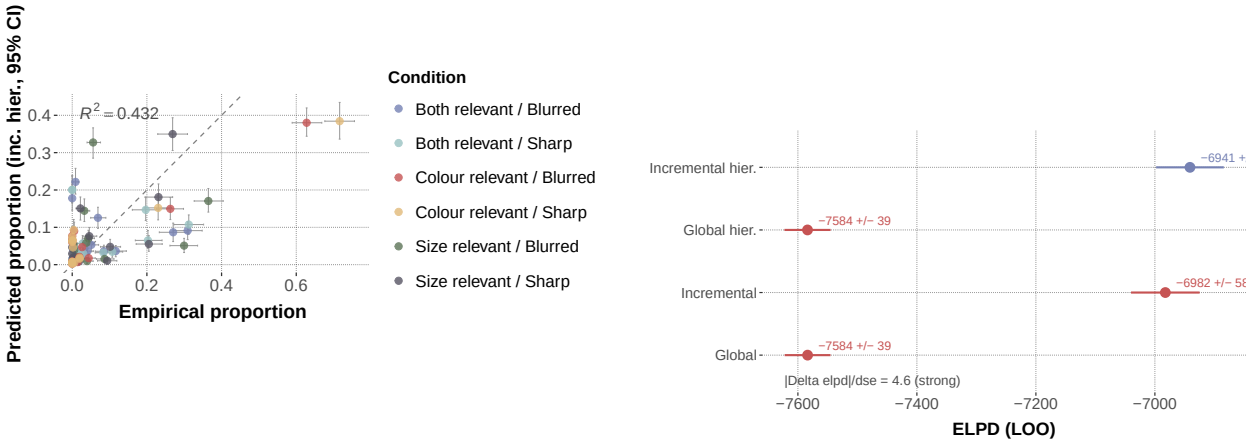

Figure 5: Left: Observed vs. predicted utterance-type proportions (per condition × utterance type) for the incremental hierarchical model, with R^2 annotation and 95% CIs. Right: LOO-ELPD comparison across four model variants; $|\Delta\text{elpd}|/\text{dse} = 4.6$ (strong evidence for the incremental hierarchical model).

3 Model

3.1 Representational primitives

State space. An **object** is a triple $o = (s, c, f)$ where $s \in [1, 10]$ is size (used to construct experiment items), $c \in \{0, 1\}$ indicates colour match, and $f \in \{0, 1\}$ indicates form match. A **context** $\mathcal{C}$ is a set of n_{obj} objects; the referent is always the largest object with $c = f = 1$.

Utterance space. Utterances are built from two semantic types: S (size) and C (colour/form; form is subsumed under colour because they share the same determinate semantics). An utterance $u = (a_1, \ldots, a_T)$ is an ordered sequence of adjectives $a_t \in \{\text{S}, \text{C}, \text{F}\}$, where $T \in \{1, 2, 3\}$ is the utterance length. Combining lengths and orderings yields 15 surface forms: {S, SC, SCF, SF, SFC, C, CS, CSF, CF, CFS, F, FS, FSC, FC, FCS}. In all analyses, F is treated identically to C.

3.2 Meaning functions

Context-dependent, threshold-based size (S) semantics. We adopt a *slack semantics* based on the Approximate Number System. A context-dependent threshold is computed as

$$\theta_k(\mathcal{C}) = x^{\text{mid}}_{\text{max}}(\mathcal{C}) - k\left(x^{\text{mid}}_{\text{max}}(\mathcal{C}) - x^{\text{mid}}_{\text{min}}(\mathcal{C})\right), \tag{1}$$

where $x^{\text{mid}}_{\text{min}}$ and $x^{\text{mid}}_{\text{max}}$ are the q_{low}- and q_{high}-quantiles of the contextual size distribution ($q_{\text{low}} = 0.2$, $q_{\text{high}} = 0.8$). The semantic value of applying S to object o is then

$$[\![\text{S}]\!](o, \mathcal{C}) = \Phi\left(\frac{s_o - \theta_k(\mathcal{C})}{w_f\sqrt{s_o^2 + \theta_k(\mathcal{C})^2}}\right), \tag{2}$$

where θ_k is the threshold from eq. (1), Φ is the standard normal CDF, and $w_f \geq 0$ is a *perceptual blur* parameter.

Colour/Form (C, F). Both colour and form have determinate denotations: an object either matches the described colour/form or it does not ($c, f \in \{0, 1\}$). Their semantic value is parameterised by a scalar $\nu \in (0, 1)$, interpretable as the probability that a feature-matching object is correctly described by the adjective—capturing residual uncertainty in the mapping from perceptual feature to word. In the slider model both C and F share a single ν; in the production model they receive separate values (ν_{colour}, ν_{form}), but both are fixed rather than inferred (see section 3.6).

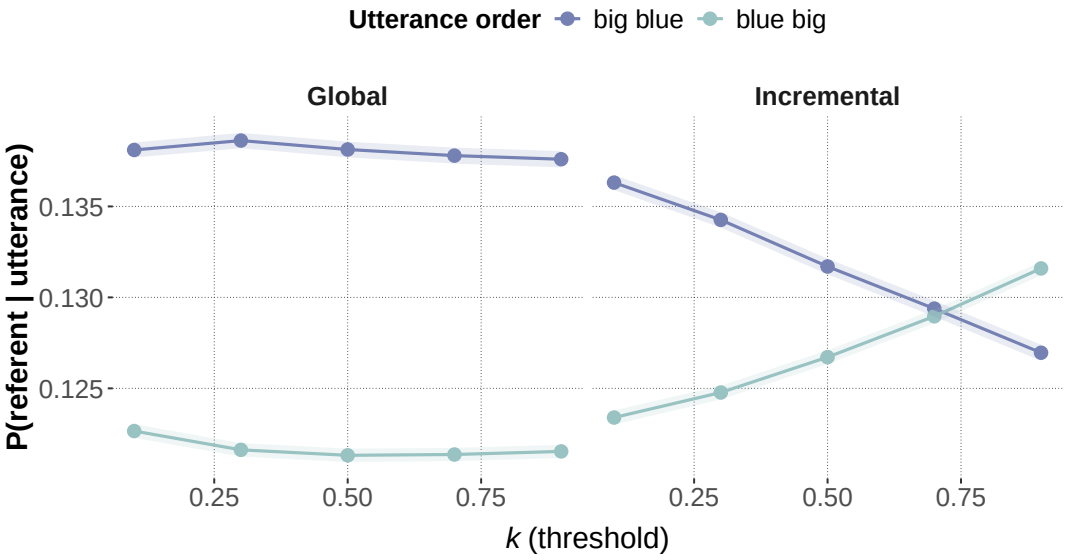

Figure 6: Effect of threshold k on communicative success (simulation, marginalised over all other parameters). Lower k (stricter threshold) reduces both orderings' success but narrows the gap between them. The incremental speaker maintains a consistent size-first advantage across k values.

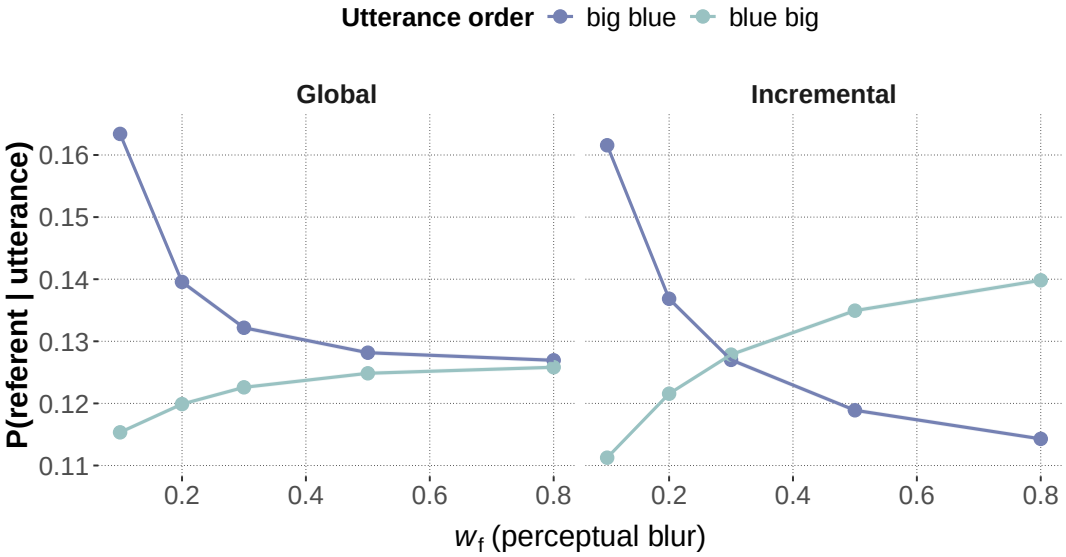

Figure 7: Effect of perceptual blur w_f on communicative success (simulation, marginalised over all other parameters). Higher w_f degrades the informativeness of size, reducing the advantage of big blue over blue big. The incremental speaker shows a larger separation between orderings than the global speaker.

3.3 Incremental listener

The incremental listener constructs utterance meanings compositionally by applying a chain-rule–based update over tokenwise semantics. Crucially, interpretation proceeds *incrementally* not only in the sense of multiplying semantic contributions, but also in the sense that each update *redefines the context* relative to which subsequent tokens are evaluated.

Chain–rule factorisation. Let $u = (w_1, \ldots, w_T)$ be an utterance and let o range over objects. The listener maintains a sequence of belief states $\{P_t(o)\}_{t=0}^{T}$, where $P_t(o)$ represents the listener's

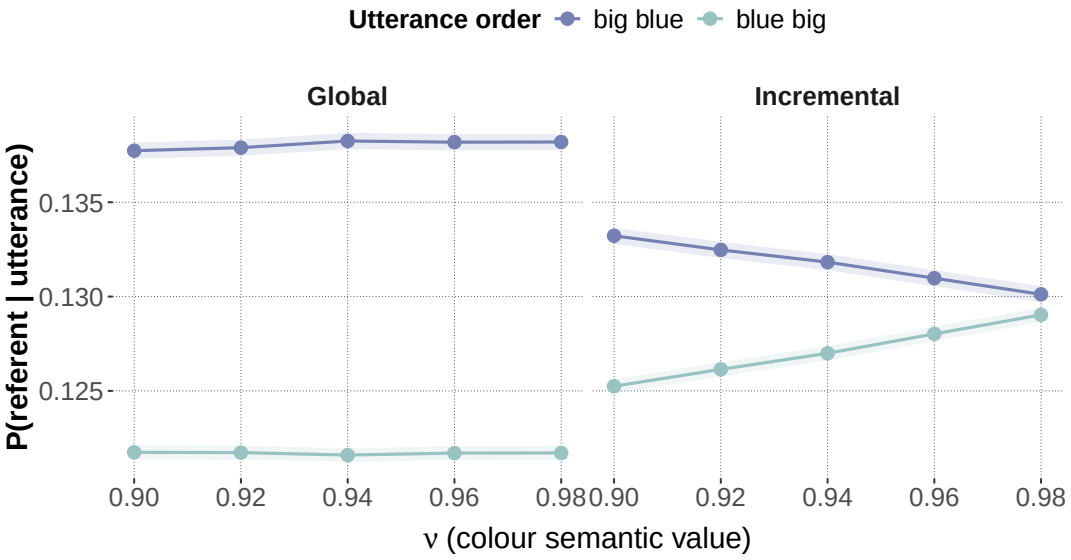

Figure 8: Effect of colour semantic value ν on communicative success (simulation, marginalised over all other parameters). Higher ν (more deterministic colour semantics) increases blue big success, narrowing the gap. The incremental speaker still favours size-first ordering across all ν values tested.

posterior belief after incorporating the suffix $(w_{t+1}, \ldots, w_T)$. Formally,

$$P_t(o) = P\big(o \mid w_{t+1}, \ldots, w_T\big), \qquad t = 0, \ldots, T, \tag{3}$$

with base case

$$P_T(o) = P_{\text{prior}}(o). \tag{4}$$

For $t = T, T-1, \ldots, 1$, beliefs are updated according to

$$P_{t-1}(o) \;\propto\; P_t(o)\, [\![w_t]\!]_{P_t}(o), \tag{5}$$

where $[\![w_t]\!]_{P_t}(o)$ is the semantic value of token w_t *evaluated relative to the current context* P_t. Normalisation yields

$$P_{t-1}(o) = \frac{P_t(o)\, [\![w_t]\!]_{P_t}(o)}{\sum_{o'} P_t(o')\, [\![w_t]\!]_{P_t}(o')}. \tag{6}$$

After all tokens have been incorporated, the listener arrives at the utterance-level posterior

$$P_0(o) = P(o \mid u), \tag{7}$$

which corresponds to the appropriate row of the listener matrix M_{listener}.

Context recursion. A key property of this formulation is that the current belief state P_t serves a *dual role*. First, it is multiplied with the semantic contribution of the next token. Second—and critically—it also determines the semantics of that token itself. Thus, the update from P_t to P_{t-1} is not merely multiplicative but *recursive*:

$$[\![w_t]\!]_{P_t} \quad \text{depends on} \quad P_t(o) \;\propto\; P_{t+1}(o)\, [\![w_{t+1}]\!]_{P_{t+1}}(o). \tag{8}$$

As a consequence, the interpretation of each token is conditioned on the entire suffix that follows it. Later words reshape the context in which earlier words are interpreted, rather than simply contributing independent semantic factors.

Ordering effects. Because certain lexical meanings—most notably size adjectives—depend on the current context distribution, changing the order of words alters the sequence of contexts $\{P_t\}$ and thereby the meanings computed at each step. This breaks the commutativity that would hold under static or Boolean semantics:

$$P(o \mid w_1, w_2) \neq P(o \mid w_2, w_1) \quad \text{in general.} \tag{9}$$

Incrementality therefore arises from two interacting sources: (*i*) chain-rule composition of tokenwise semantics, and (*ii*) recursive context dependence. Together, these properties yield a genuinely incremental listener whose interpretations are sensitive to word order in a way that cannot be reduced to a global, order-insensitive update.

3.4 Utterance prior (cost)

We incorporate two cost signals into the utterance prior, depending on the dataset:

(i) **Slider rating — subjectivity bias:** an additive penalty favouring size-first (S-initial) orderings, captured by a bias parameter b.

(ii) **Production — LM-derived cost:** $\text{cost}_{\text{LM}}(u) = -\beta \log P_{\text{LM}}(u)$, where P_{LM} is a GPT-based prior over the 15 utterances and β controls sensitivity. This cost implicitly captures frequency and length biases: shorter, high-frequency forms (*e.g.* S) are cheaper.

3.5 Speaker models

3.5.1 Global speaker

The global speaker computes a distribution over complete utterances by applying a softmax to utterance-level utilities:

$$S_{\text{gb}}(u \mid o) = \frac{\exp\big(U(o,u)\big)}{\sum_{u'} \exp\big(U(o,u')\big)}. \tag{10}$$

The utility of an utterance u for an intended referent o^* is defined as

$$U(o^*, u) = \alpha \log L(o^* \mid u) + \beta \log P_{\text{LM}}(u), \tag{11}$$

where α is the RSA rationality parameter governing sensitivity to informativeness, and $\beta \geq 0$ scales the contribution of the language-model prior, which encodes utterance frequency and length biases (higher β favours high-probability LM forms).

Slider variant. For the slider-rating models, which compare only two utterances (big blue vs. blue big), the utility takes the form

$$U(o, u) = \alpha\big(\log L(o \mid u) - b_u\big), \tag{12}$$

where b_u is the subjectivity bias ($b_u = 0$ for the size-first ordering, $b_u = b$ for the reverse ordering).

For each intended object o_j, this yields a full distribution over utterances. The resulting probability vector constitutes the jth row of the global speaker matrix M_{gsp}. In the experiment, the speaker always refers to a fixed target referent o_r. Under our indexing scheme, the target referent corresponds to index 1. Accordingly, the model prediction entering the likelihood is

$$\mathbf{p} = (p_1, \ldots, p_{15}) = P(u \mid o_r), \tag{13}$$

which is given by the first row of M_{gsp}.

3.5.2 Incremental RSA speaker

Chain-rule factorisation. For a fixed intended referent o^*, the incremental pragmatic speaker constructs utterances token by token. Formally, utterance production obeys the chain rule:

$$S_{\text{inc}}(u \mid o^*) = \prod_{t=1}^{T} S_t(w_t \mid u_{<t},\, o^*)\,, \qquad u_{<t} = (w_1, \ldots, w_{t-1}), \tag{14}$$

where $S_t(\cdot)$ denotes the speaker's local decision rule at step t.

Local RSA utility. At each step, the speaker evaluates possible next tokens a solely in terms of their informativeness for the intended referent. The local decision rule is

$$S_t(a \mid u_{<t}, o^*) \;\propto\; \big[L(o^* \mid u_{<t} + a)\big]^{\alpha}, \tag{15}$$

where $L(o^* \mid u)$ is the incremental literal listener defined over context-dependent semantics, and α is the RSA rationality parameter. Thus, at the token level the incremental speaker reasons only about *informativeness*: utterance cost does not enter into the stepwise utility. Costs derived from the external language model apply only at the level of complete utterances, as described in the utterance-cost subsection.

Composition of utterance-level cost and incremental pragmatics. Combining the chain-rule factorisation with utterance-level cost yields the full production probability for an utterance $u = (w_1, \ldots, w_T)$:

$$S_{\text{inc}}(u \mid o^*) = P_{\text{LM}}(u)^{\beta} \cdot \prod_{t=1}^{T} \frac{\big[L(o^* \mid u_{<t} + w_t)\big]^{\alpha}}{Z_t(u_{<t}, o^*)}, \tag{16}$$

where $P_{\text{LM}}(u)$ is the language-model probability of the complete utterance and

$$Z_t(u_{<t}, o^*) = \sum_{a} \big[L(o^* \mid u_{<t} + a)\big]^{\alpha} \tag{17}$$

is the local normalisation constant ensuring that $S_t(\cdot \mid u_{<t}, o^*)$ defines a proper probability distribution over possible next tokens.

This formulation makes explicit that utterance cost enters the model only once, as a global multiplicative factor, while pragmatic reasoning operates incrementally through successive evaluations of local informativeness. As a result, ordering effects in the incremental speaker arise from sequential semantic context update rather than from token-level cost or language-model biases.

Intuition: why incremental and global speakers are not equivalent. Although both the global and incremental speakers assign probabilities over the same utterance space, they are not in general equivalent. The key difference lies in how pragmatic reasoning is applied. In the global speaker, pragmatic reasoning is applied once, at the level of complete utterances. In the incremental speaker, pragmatic reasoning is applied repeatedly and locally, at each step of utterance construction, relative to an evolving semantic context.

Formally, in the global speaker, utterance utility is computed as

$$U_{\text{gsp}}(o^*, u) = \alpha \log L(o^* \mid u) + \beta \log P_{\text{LM}}(u), \tag{18}$$

and normalised once over complete utterances. In contrast, the incremental speaker decomposes pragmatic reasoning across token positions:

$$S_{\text{inc}}(u \mid o^*) = P_{\text{LM}}(u)^{\beta} \prod_{t=1}^{T} \frac{\big[L(o^* \mid u_{<t} + w_t)\big]^{\alpha}}{Z_t(u_{<t}, o^*)}. \tag{19}$$

Because the listener $L(o^* \mid u)$ depends on context-dependent semantics, the marginal contribution of a word differs depending on when it is evaluated. As a result, the product of locally optimal pragmatic choices generally does not coincide with the globally optimal utterance-level choice. Thus,

$$S_{\text{inc}}(u \mid o^*) \neq S_{\text{gsp}}(u \mid o^*) \quad \text{in general,} \tag{20}$$

even when both models share the same language-model cost and rationality parameter.

Intuitively, the incremental speaker optimises *incremental informativeness*, while the global speaker optimises *total informativeness*. When semantics is context-dependent—as in the present model—these two criteria diverge, yielding genuine ordering effects that cannot be reduced to utterance-level cost differences.

3.6 Model variants & parameterisation

Crossing the two speaker models (S_{G} vs. S_{I}) with two levels of participant-level pooling yields four model variants per dataset, compared via LOO-CV. The semantic parameters k, w_f, and ν are held fixed at theoretically motivated values (see below); inference targets the pragmatic and observation-level parameters.

Slider-rating models. The slider models compare only two utterances (big blue vs. blue big), using a subjectivity bias as the cost term and a Zero-One-Inflated Beta (ZOIB) observation model.

Model	Speaker	Participant pooling	Inferred parameters
Global, pooled	S_{G}	Single parameter set across all participants	α, b, σ
Global, hierarchical	S_{G}	Per-participant additive intercepts δ_i	α, b, σ, τ, δ_i
Incremental, pooled	S_{I}	Single parameter set across all participants	α, b, σ
Incremental, hierarchical	S_{I}	Per-participant additive intercepts δ_i	α, b, σ, τ, δ_i

Production models. The production models evaluate all 15 utterances with LM-derived costs. The hierarchical variants add per-participant deviations on α.

Model	Speaker	Participant pooling	Inferred parameters
Global, pooled	S_{G}	Single parameter set across all participants	α, β
Global, hierarchical	S_{G}	Per-participant deviations on α	α, β, τ, δ_i
Incremental, pooled	S_{I}	Single parameter set across all participants	α, β
Incremental, hierarchical	S_{I}	Per-participant deviations on α	α, β, τ, δ_i

Fixed semantic parameters (shared across all model variants): $k = 0.5$, $w_f = 1.0$, $\nu_{\text{colour}} = 0.8$ (slider and production), $\nu_{\text{form}} = 0.7$ (production only). Although not inferred, these parameters play a central role in shaping the model's predictions; figs. 6 to 8 illustrate their effects via the simulation (section 4).

Parameter glossary.

Parameter	Support	Interpretation
Inferred		
α	$\mathbb{R}_{>0}$	Pragmatic rationality: sharpness of the speaker's softmax over utterances
b	$\mathbb{R}_{>0}$	Subjectivity bias favouring S-initial orderings (slider models only)
σ	$\mathbb{R}_{>0}$	ZOIB dispersion controlling the spread of predicted slider ratings (slider models only)
β	$\mathbb{R}_{>0}$	Sensitivity to LM-derived utterance cost (production models only)
τ	$\mathbb{R}_{>0}$	SD of participant-level random effects (hierarchical variants only)
δ_i	$\mathbb{R}$	Per-participant intercept or deviation (hierarchical variants only)
Fixed (not inferred; effects shown in figs. 6 to 8)		
k	$(0, 1)$	Threshold position within the contextual size range; lower k = stricter threshold (fig. 6)
w_f	$\mathbb{R}_{\geq 0}$	Perceptual blur in the Weber-fraction size comparison; higher w_f = noisier size semantics (fig. 7)
ν	$(0, 1)$	Colour/form semantic value: probability that a feature-matching object is truly described by the adjective; higher ν narrows the size-first advantage (fig. 8)

Discussion: Challenges and experiences in fitting the models

- **Computational efficiency.** Making inference tractable required systematic vectorisation of data and parameters (leveraging JAX's batch semantics), JIT compilation of the entire model graph, and precomputing all quantities that do not depend on MCMC samples—in particular, marginalising out stochastic dependencies that could otherwise be recomputed at every leapfrog step.
- **Slow and stuck chains; chain diagnostics.** NUTS chains were frequently slow or got stuck in low-probability regions, making the usual workflow of running many short diagnostic chains prohibitively expensive. This motivated a dedicated posterior-analysis pipeline to detect problems early: visual inspection of posterior distributions for bad shapes, number of divergences, cross-chain disagreement, high autocorrelation within chains, and strong parameter correlations. Defining what counts as "successful inference" (*e.g.* by inspecting by-condition posterior predictives) helps a lot.
- **Slider rating data: bathtub-shaped distribution.** Slider ratings pile up at 0 and 1 (boundary values), violating standard continuous likelihood assumptions. We adopted a Zero-One-Inflated Beta (ZOIB) likelihood that explicitly models boundary probability mass via mixture components—a point mass at 0 with probability π_0, a point mass at 1 with probability π_1, and a Beta distribution for interior values with probability $1 - \pi_0 - \pi_1$.
- **Production data: degenerate and context-dependent distributions.** The free-production data exhibit highly concentrated, context-dependent categorical distributions—many utterance types receive near-zero counts in some conditions—making the full likelihood surface very steep and difficult to navigate. Our solution was to fix the semantic parameters (k, w_f, ν) at shared values across both datasets and infer only the pragmatic parameters (α, β) from the production data.

4 Simulation

4.1 Goals

The simulation has three purposes:

(1) Establish that the incremental speaker *generates* ordering preferences, not just accommodates them, across a broad range of contexts.

(2) Map out which parameter region produces a qualitative *big-before-colour* preference and compare to a *colour-before-big* region.

(3) Assess robustness: does the ordering preference persist under varying context size (n_{obj}), size-distribution spread (σ_{spread}), and perceptual blur (w_f)?

4.2 Design

Parameter	**Values swept**	**# levels**
Speaker model	Incremental, Global	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν (colour semantic value)	0.90, 0.92, 0.94, 0.96, 0.98	5
k (threshold)	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f (perceptual blur)	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination		1000
Total rows		**6,000,000**

The parameter σ_{spread} controls the standard deviation of the truncated normal from which object sizes are drawn within a context. A small value (e.g. $\sigma_{\text{spread}} = 2$) produces *blurred* contexts in which objects are similar in size, making size adjectives less informative; a large value (e.g. $\sigma_{\text{spread}} = 15$) yields *sharp* contexts with well-separated sizes, making size adjectives highly informative.

Each simulated trial computes communicative success $p(o_{\text{ref}} \mid u)$ for two key orderings (big blue > blue big) and records which ordering the speaker model assigns higher probability.

4.3 Results

Discussion: Simulation findings

- Both speaker types generate a size-first ordering advantage across the majority of the parameter space, demonstrating that the emergence of ordering preferences is not exclusive to the incremental mechanism.
- The global speaker derives its ordering asymmetry solely from context-dependent, sequential-update compositional semantics—i.e., the way the literal listener composes meaning by updating beliefs object-by-object. This is the *minimal mechanism*: it requires no production-level incrementality, only a compositional semantics that is sensitive to contextual informativity. This is compatible with previous listener-oriented communicative-efficiency accounts of subjectivity-based ordering preferences. While those accounts relied on non-agent simulations, our explicit RSA model captures, extends, and confirms their core insight within a fully agent-based framework.
- The incremental speaker amplifies this effect and additionally produces qualitatively distinct predictions (e.g., a *reversal* to colour-first preference in blurred, small-context conditions; fig. 9), suggesting that the incremental production mechanism contributes an additional, separable source of ordering preferences beyond the minimal compositional-semantics baseline.

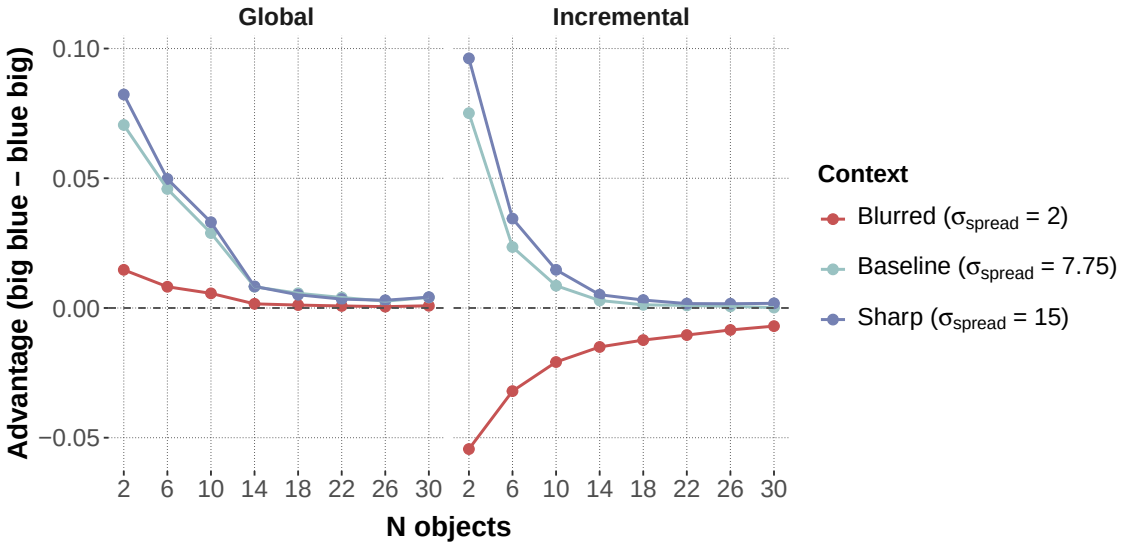

Figure 9: Size-first advantage (`big blue` − `blue big`) as a function of context size n_{obj} and context sharpness σ_{spread}, faceted by speaker type. Both speakers show a positive advantage in sharp contexts that diminishes with increasing context size. Crucially, the incremental speaker exhibits a *negative* advantage (colour-first preference) in blurred contexts with few objects, a pattern absent from the global speaker.

5 General Discussion

5.1 Summary

We presented an RSA account of adjective ordering with two speaker variants—global and incremental—embedded in a shared context-sensitive compositional semantics. The empirical motivation for this comparison was that adjective ordering has been explained from at least two partially overlapping perspectives in the recent literature: a *subjectivity-based* account, in which less subjective adjectives are more reliable under restrictive composition, and a broader *communicative-efficiency* account, in which adjective order is shaped by how efficiently an utterance guides reference resolution in context. Our experiments and simulations were designed to ask how far the minimal compositional-semantics mechanism already goes, and what additional explanatory value is gained by modeling production itself as incremental.

Discussion: Key findings

- **The baseline ordering effect does not require an incremental speaker.** In the simulations, even the global speaker—which evaluates complete utterances rather than producing them word by word—shows a robust size-first preference across much of parameter space. This result is important because it aligns our model with prior subjectivity-based and incremental-semantic-restriction accounts: a substantial part of the canonical ordering asymmetry can already be derived from context-sensitive restrictive composition (**???**).
- **Incremental production nevertheless adds explanatory value.** The incremental speaker does not merely reproduce the baseline effect. It amplifies the ordering asymmetry and, in some parts of parameter space, generates qualitatively distinct predictions, including colour-first reversals in blurred, low-contrast contexts (fig. 9). This suggests that production-level incrementality is best understood not as the sole source of adjective-ordering preferences, but as an additional mechanism that modulates a more basic efficiency effect.
- **The human data support a discriminatory-strength account of ordering in context.** Across the two experiments, ordering preferences vary systematically with which

property is most useful for identifying the target. This connects our findings directly to the discriminatory-efficiency hypothesis (**?**) and to the broader incremental-efficiency literature on reference and modifier use (**????**).

- **The best-fitting model is the incremental hierarchical model.** Although the global model captures the minimal compositional-semantic mechanism, the incremental hierarchical model provides the strongest quantitative fit to both slider and production data. On our interpretation, this means that the basic direction of adjective-ordering preferences follows from efficient restrictive composition, while the detailed pattern of human behavior is better captured when production is modeled as an online, stepwise decision process.

5.2 From Subjectivity to Communicative Efficiency

One contribution of the present results is to sharpen the relation between subjectivity-based and communicative-efficiency approaches to adjective ordering. The subjectivity literature shows that less subjective adjectives tend to occur closer to the noun, and explains this in terms of semantic reliability under hierarchical composition (**???**). Our simulations support this general idea: once adjective meanings are interpreted relative to restricted comparison classes, a global speaker already derives the baseline ordering asymmetry. In this sense, the current results are cumulative with the subjectivity literature rather than revisionary.

At the same time, the experiments show that adjective ordering in referential communication is not exhausted by static adjective-class preferences. What matters behaviorally is not only whether an adjective is generally more or less subjective, but also whether it is *currently useful* for discriminating the target from its competitors. This is why the present findings fit naturally with Fukumura's discriminatory-efficiency hypothesis and with Rubio-Fernández's broader claim that the efficiency of a referential expression must be evaluated incrementally, over the full visual context. Under this perspective, subjectivity provides a baseline tendency, while discriminatory strength explains how that tendency is modulated, strengthened, or sometimes locally counteracted by the current referential scene.

5.3 Do We Need an Incremental Speaker?

The answer suggested by the present results is nuanced. If the question is whether production-level incrementality is necessary to derive *any* adjective-ordering preference, the answer is no. The global speaker already captures a substantial part of the phenomenon, because the ordering asymmetry can emerge from context-sensitive compositional semantics alone. This is the minimal mechanism.

If the question is instead whether production-level incrementality improves the account of *human behavior*, the answer appears to be yes. The incremental model fits both datasets better and predicts context-sensitive patterns that are harder to obtain from the global speaker alone. The relevant contrast is therefore not between "incremental speaker" and "no theory," but between two levels of explanation:

(a) a minimal communicative account grounded in restrictive composition, and

(b) a richer online-production account in which speakers optimize adjective choice step by step.

This division of labor also clarifies the broader theoretical contribution of the paper. We are not arguing that adjective ordering is reducible to a single mechanism. Rather, we show how stable lexical-semantic asymmetries and context-specific discriminatory pressures can be integrated within a single probabilistic framework, and we provide evidence that online production planning helps explain the finer structure of human ordering behavior.

5.4 Broader Implications

Taken together, the results support a view of adjective ordering preferences as graded, context-sensitive adaptations for successful reference resolution. This perspective connects three strands of research that are often discussed separately. First, it preserves the central insight of subjectivity-based theories: semantically more reliable adjectives are favored in structurally inner positions. Second, it incorporates the communicative-efficiency literature, in which modifier use is sensitive to discriminatory value and visual search. Third, it embeds these pressures in an explicit incremental pragmatic model that makes quantitative predictions about both judgment and production.

More broadly, this means that adjective ordering preferences should not be treated as the output of a fixed hierarchy alone, nor as the product of a single lexical predictor. They are better understood as efficient solutions to a referential communication problem in which lexical-semantic properties, context structure, and online utterance planning interact.

Discussion: Outlook

- **Ablation priority.** The most informative next step is to disentangle context-recursive semantics from production-level incrementality by adding static-semantics ablations (section 4; see also `further_model_comparison_idea.tex`).
- **Theoretical extension.** A useful longer-term comparison is with non-RSA discriminatory models such as emitter-suppressor accounts, which would test whether listener-based pragmatic inference is necessary beyond direct distractor suppression.
- **Broader framing.** The paper is best positioned at the intersection of pragmatics, psycholinguistics, and cognitive science: it links adjective-ordering preferences to efficient, incremental reference resolution rather than to a language-specific ordering template alone.

References

[TODO: Add references]